SURIYA GANAPATHI K

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

AI Engineer at DoodleBlue Innovations who has shipped 4+ production AI applications, including an OCR-based document processing platform for the Nippon client and a multilingual IVR voice agent for a bank handling 500+ calls per day. BTech in Artificial Intelligence and Data Science (2025). Builds LLM-powered pipelines, RAG architectures, IVR-integrated platforms, and intelligent document processing with Google Gemini, OpenAI, Claude, Groq, and Sarvam AI, using Python, SQL, FastAPI, and LangGraph across NLP, computer vision, and predictive analytics.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS, JavaScript

Generative AI & Agentic AI: LangChain, LangGraph, LangSmith, Langfuse, RAG Pipelines, Prompt Engineering, Agentic Workflows, Multi-Agent Systems, Vector Databases

LLMs & AI Platforms: OpenAI, Google Gemini, Anthropic Claude (Sonnet, Haiku), Groq, Sarvam AI, AWS Bedrock

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, NLP, RNN, Sentiment Analysis, Logistic Regression, Decision Trees, Random Forest, K-Means, PCA

Databases & Vector Stores: MongoDB (Vector Search), MySQL, PostgreSQL, FAISS, ChromaDB, Pinecone

Frameworks & Libraries: Flask, FastAPI, Streamlit, Pandas, PyMuPDF, EasyOCR

Data Visualization & BI: Power BI, MS Excel

Tools & Platforms: Git/GitHub, Postman, Swagger

WORK EXPERIENCE

AI Engineer – DoodleBlue Innovations | Feb 2026 – Present
Full-Time | Chennai, Tamil Nadu

- Developed and maintained 4+ production AI applications involving RAG, intelligent document processing, and conversational AI systems for enterprise clients, including an OCR-based document processing platform for the Nippon client, cutting manual processing effort by ~70%.
- Delivered Gemini, OpenAI, and Groq API-based solutions across multiple client engagements, covering document intelligence, retrieval systems, and multilingual workflows in 4 languages (Hindi, Tamil, English, Japanese).
- Worked with the team to design and deploy AI backends that connect cloud APIs, handle multiple languages and meet regulatory requirements.

AI/ML Intern – DoodleBlue Innovations | Jul 2025 – Jan 2026
Internship | Chennai, Tamil Nadu

- Built 3 end-to-end AI systems: an IVR-based debt collection platform (Gemini, Sarvam AI, Sarvam telephony), a RAG chatbot (OpenAI + MongoDB), and an IDP & translation platform (Gemini, Groq, EasyOCR).
- Worked on every stage, from data ingestion and model integration to deployment and later improvements, earning conversion to full-time after 7 months.

AI & Data Science Intern – EduFyi Tech Solutions | Jul 2025 – Sep 2025
Internship | Remote

- Built predictive models and did exploratory data analysis on real-world datasets.
- Used Python, Scikit-learn and data visualization tools day to day.

Big Data Intern – Code Bind Technologies | 2024 | 15 Days
Internship

- Designed a data warehouse architecture for an e-commerce platform using Big Data technologies, applying ETL pipeline concepts to build dimensional models optimized for BI reporting.

PROJECTS

Retail & Fibre Analytics and Planning Agent
Built a multi-agent conversational AI system across 3 domains (Retail, Fibre, and B2C) using Python, FastAPI, LangGraph, and AWS Bedrock (Claude Sonnet 4.6 and Haiku), with 2 agents: an Insights agent that answers business questions and a Planning agent that suggests next steps from verified data. Connected the agents to read-only PostgreSQL analytics and Pinecone semantic metric search so answers follow approved business definitions. Built the backend platform with LLM model routing, swappable model-provider adapters, privacy controls, run tracking, and evaluation checks before release.

AI-Powered Marketing Content Platform
Designed and developed an agentic AI marketing automation platform using LangChain and LangGraph to orchestrate multi-step content generation and campaign workflows. Built stateful agent architectures for personalized text and image creation, semantic retrieval, and multi-turn context management, so content could be generated and sent across several customer channels.

AI Debt Collection Platform
Developed an AI-powered debt collection platform for a banking client integrating Google Gemini, Sarvam AI, and Sarvam telephony APIs, automating borrower interactions and supporting multilingual voice conversations in Hindi, Tamil, and English across 100+ concurrent calls and 500+ calls per day. Implemented real-time sentiment analysis, dynamic conversation adaptation, and rule-based guardrails so calls stayed within RBI rules.

RAG Document Search Project
Built a Retrieval-Augmented Generation (RAG) application using LangChain for intelligent document search and context-aware question answering. Implemented a complete retrieval pipeline covering document ingestion, text chunking, vector embedding generation, and FAISS-based semantic similarity search, with LangGraph-inspired workflow orchestration enabling users to query enterprise knowledge bases in natural language.

Intelligent Document Processing (IDP) & Translation Platform
Built an OCR-driven IDP pipeline for the Nippon client using a custom parser factory (PyMuPDF, Pandas, EasyOCR) with Google Gemini and Groq LLMs to automate extraction of structured fields from digital and scanned documents, reducing manual processing effort by approximately 70% and achieving ~85% extraction accuracy on a 20-document test set. Integrated zero-shot classification, schema-based field extraction, and English-to-Japanese translation with custom glossary mapping, delivering a multi-format document intelligence solution across 3 domains: finance, legal, and logistics.

2024 Indian General Elections Analysis – Power BI Dashboard
Built an interactive Power BI dashboard on the 2024 Indian General Elections with four views: Overview, State Demographics, Constituency Analysis and State-wise Performance. Modelled the data in SQL to look at voter turnout and how results differed across states and constituencies.

PCOS Detection Using Machine Learning (RNN)
Built a full-stack PCOS prediction web application using Flask, MySQL, and HTML/CSS/JavaScript, enabling users to input health parameters for real-time analysis. Trained and optimized an RNN model on medical datasets to classify PCOS risk, with a simple UI that shows the prediction for early health screening.

EDUCATION

B.Tech – Artificial Intelligence and Data Science | 2021 – 2025
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Tamil Nadu

Higher Secondary (Class XII) | 2019 – 2021
New Prince Matriculation Higher Secondary School, Tamil Nadu

Secondary (Class X) | 2018 – 2019
Dayananda Anglo Vedic (D.A.V.) School, Tamil Nadu

CERTIFICATIONS

- Power BI – Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College
- Python Programming – Top Engineers (Workshop)
- Ultimate RAG Bootcamp Using LangChain, LangGraph & LangSmith – Udemy
- Complete Agentic AI Bootcamp With LangGraph and LangChain – Udemy